The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Multilayer Cyber-Physical Decision Symmetries and Slack Interlocks

We formalize the symmetric decentralization of the electrical transmission network by mapping the power grid into a functional, multi-layer conceptual decision hierarchy.

3.1 Mesarovic Complexity Layers and Vertical Response Vectors

The control topology splits across three distinct functional complexity layers, eliminating top-down monolithic dependencies:

- **Layer 1 (Direct Control Layer):** Operates on a sub-millisecond to second interval. It uses vectorized numerical optimization and continuous DAE solvers to regulate immediate feedback.
- **Layer 2 (Adaptation Layer):** Operates on a minute to hourly interval. It utilizes statistical pattern recognition and logical inductive learning forests to identify localized system models.

- **Layer 3 (Self-Organizing Layer):** Operates on a weekly to monthly timeline. It executes management-by-exception heuristics to modify macro-operating strategies upon global performance deterioration.

Let $\mathcal{S}_{\text{slack}}$ define a continuous vector interlock managing the boundary constraints between Layer 1 transient responses and Layer 2 Area Generation Control (AGC) commitments under PJM Manual 12 Balancing Operations:

$$\text{Minimize} \quad \mathcal{J} = \int_{t}^{t+T} \left(\boldsymbol{\xi}_i(t) - \boldsymbol{\xi}_{\text{nominal}}\right)^2 dt + \boldsymbol{s}_{\text{slack}}^T \boldsymbol{W} \boldsymbol{s}_{\text{slack}} \tag{2}$$

The slack variables $\boldsymbol{s}_{\text{slack}}$ prevent logical contradictions between high-speed inverter phase-slip damping and slow, hourly economic dispatch limits, certifying local stability without forcing aggregators to disclose internal control laws or proprietary parameters.

3.2 Virtual LAN Grid Partitioning Invariants

To optimize security and maintain firewall integrity, the macro-grid topology is partitioned into a set of small, decentralized virtual area local area networks (LANs). Instead of a single area with M physical generators, the graph is divided into M independent virtual zones containing a single generation asset or localized renewable cluster each. High-speed Graph Neural Networks (GNNs) compute the inter-LAN boundary exchange paths across the partitioned adjacency matrix $\boldsymbol{A}_{\text{partition}}$:

$$\boldsymbol{A}_{\text{partition}} = \bigoplus_{k=1}^{M} \boldsymbol{A}_{\text{LAN},k} \in \mathbb{R}^{N \times N} \tag{3}$$

Each virtual area network connects to the existing power data grid through a single secure gateway interface, eliminating individual node vulnerabilities and allowing the Black-Hunter-Pan dialogue poset function (Scounter $\succ$ Sarg $\succ$ Sconcede $\succ$ Sclose) to resolve turn-taking deconflictions locally during consensus rounds.